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Response to the White House Request for Information on the Development of an Artificial Intelligence (AI) Action Plan:

The Role of Edge AI in U.S. Competitiveness, Innovation, and Security

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Statement

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Introduction: The Importance of Edge AI in the National AI Policy

Artificial Intelligence (AI) has become the defining technological race of the 21st century, with transformative impacts on economic growth, national security, and global competitiveness. While much of the public discourse on AI focuses on cloud-based models and data centers, Edge AI—where AI computation occurs near data sources rather than in centralized cloud servers—represents a fundamental shift in AI deployment and innovation. The NSF AI Institute for Edge Computing, Athena, is at the forefront of advancing Edge AI research, fostering interdisciplinary collaboration among academia, industry, and government to develop energy-efficient, real-time, and privacy-preserving AI solutions at the edge. This response outlines critical policy considerations necessary to ensure the United States maintains its leadership in AI by prioritizing investment in AI infrastructure, workforce development, international competitiveness, and security.

1. The Need for Robust AI Infrastructure Investments

The U.S. must significantly increase investments in AI infrastructure, particularly in Edge AI, to keep pace with global competitors. While tech giants in the U.S. focus on large-scale cloud AI, other nations have aggressively expanded their capabilities in AI hardware, edge computing, and sovereign AI models, controlling access to AI applications for end users.

Some countries have launched advanced AI models with greater autonomy from chip and cloud providers, ensuring their technological sovereignty by investing heavily in custom AI semiconductors and Edge AI deployments. These initiatives are designed to reduce dependency on U.S.-based AI technology, signaling a shift in the global AI power balance.

To maintain its leadership, the U.S. must:

1. Fund Edge AI-specific research that prioritizes low-latency, energy-efficient AI models.
2. Expand semiconductor investments, particularly in custom AI accelerators optimized for Edge AI.
3. Enhance domestic AI infrastructure by integrating AI-driven networking, edge datacenters, and federated AI systems across industries.

Without a focused policy effort, the U.S. risks falling behind as global players advance self-sufficient AI ecosystems that are independent of U.S.-based AI models and cloud infrastructure.

2. AI Workforce Development: Addressing the Talent Gap

The U.S. faces a growing shortage of AI talent, particularly in AI hardware, Edge AI, and AI Infrastructure. The AI race is not just about technology; it is about who develops, deploys, and governs AI systems.

Current Challenges:

1. The demand for AI engineers and AI-embedded hardware specialists far exceeds the supply.
2. Visa and immigration restrictions, along with geopolitical tensions, have limited the flow of top AI researchers to the U.S.
3. Gaps in STEM education, declining federal funding for university research, and uneven AI literacy create barriers to cultivating the AI workforce.

Policy Recommendations:

1. Increase federal support for AI research and education programs at all levels, particularly in AI hardware, Edge AI, and AI infrastructure in general.
2. Expand public and private AI training initiatives with industry-sponsored apprenticeships.
3. Create targeted immigration policies to attract top-tier AI talent, ensuring the U.S. retains its intellectual edge in AI research and deployment.

Without strategic investment in AI workforce development, the U.S. risks a critical skills gap that could slow AI innovation and adoption.

3. Global AI Competition: Maintaining U.S. Leadership

The global AI competition is intensifying, with other nations making strategic, long-term investments in the field. Unlike the U.S. AI ecosystem, which is primarily driven by the private sector, these nations often take a government-led approach to investing in AI research and shaping AI policy.

Some countries have aggressively pursued AI chip sovereignty, reducing their reliance on hardware designed by U.S. companies such as NVIDIA, AMD, Intel, and Qualcomm. Instead, they are building AI ecosystems that operate independently of the U.S. and its allies. Others have prioritized privacy-first, regulated AI models that align with local data governance laws, further reducing dependence on U.S. technology providers.

Policy Recommendations:

1. Ensure sustained federal investment in AI to match or exceed global competitors.
2. Foster AI infrastructure independence, particularly in AI computing centers, AI computing chips, and edge AI devices.
3. Encourage international AI collaborations while protecting against technology leakage to adversarial nations.

Without decisive action, the U.S. risks losing its competitive edge in AI research, deployment, and commercialization.

4. AI Security & National Defense: The Role of Edge AI

Security remains one of the most pressing challenges in AI governance, especially in foundational AI infrastructure such as Edge AI.

The U.S. Must Prioritize Secure Edge AI:

1. Edge AI minimizes attack surfaces by reducing data exposure to vulnerabilities inherent in centralized cloud-based AI systems.
2. Federated learning, trusted AI hardware, and other edge security technologies establish a robust framework for decentralized AI governance, mitigating risks of adversarial AI exploitation.

Policy Recommendations:

1. Establish mandatory AI security standards for Edge AI deployment across critical AI infrastructure.
2. Increase funding for AI research in cybersecurity, focusing on adversarial AI detection and mitigation in edge applications.
3. Strengthen defense partnerships with academia to accelerate the adoption of secure AI in national security applications.

Without a secure AI infrastructure, U.S. national security is at risk in an era of autonomous cyber warfare and AI-powered misinformation campaigns.

Conclusion: A Call for Action

The United States stands at a critical juncture in AI policy. To maintain its global leadership, the nation must prioritize:

- Robust AI infrastructure investments, particularly in Edge AI.
- AI workforce development to close the talent gap.
- Strategic responses to global AI competition.
- Security-first AI policies to safeguard national interests.

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Biography of Yiran Chen

Prof. Yiran Chen is a world-renowned AI researcher and a recognized authority in AI policy discussions. As the director of Athena, he has provided expert insights on AI governance, workforce development, and security through government panels and committees, mainstream media interviews with outlets such as *BBC*, *Bloomberg*, *The New York Times*, *Reuters*, and *CNA*, as well as at academic conferences.

Introduction of Athena

The NSF AI Institute for Edge Computing (Athena) is dedicated to transforming modern edge computing systems by advancing AI foundations, computing paradigms, and applications from a novel computing perspective. Led by Duke University, Athena brings together a multidisciplinary team from eight leading universities, spanning fields such as electrical engineering, computer science, healthcare, education, psychology, political science, and law. Its research focuses on three core pillars—Edge Computing Systems, Collaborative Extended Reality (XR), and Situational Awareness and Autonomy—all grounded in Foundational AI. Athena's innovations aim to enhance the heterogeneity, efficiency, functionality, and trustworthiness (HEFT) of AI solutions in edge computing. Beyond research, Athena is committed to education, workforce development, and industry collaboration, ensuring that its advancements benefit both the public and private sectors while fostering ethical AI practices.